GOOGLE CLOUD AI BUILDER CUP 2026 · JAPAC

Kairos AI

An expert reliability engineer inside every power device.

Autonomous, remote fault diagnosis and self-healing for critical power infrastructure — catching the moment to act, before downtime.

Theme · Manufacturing

Gemini 2.5 Flash · Vertex AI

Deployed on Cloud Run

Live: kairos-ai-498775210896.asia-south1.run.app · github.com/puneetpahuja-eaton/kairos-ai

THE PROBLEM — AND OUR ANSWER

Today, finding a fault needs a human on site

THE PROBLEM

A fault is only caught when someone notices an alarm, reports it, and waits for an engineer to travel to site. Downtime compounds — and expert engineers are scarce and can't be everywhere.

- **No root-cause intelligence —**
Tools only alert; they don't explain why.
- **No prediction or correlation —**
Cascading alarms hide the true cause.
- **No closed-loop remediation —**
Only ~10% (alerting) is automated; 90% is manual.

KAIROS AI

Puts an expert reliability engineer inside every device — learning it from its own documents, watching it remotely 24/7, and closing the loop from detection to verified fix.

Learn
— Ingests datasheets & logs (RAG on Vertex AI).

Monitor
— Watches live telemetry & event logs remotely.

Diagnose
— Plain-English root cause, cited from the docs.

Act & Verify
— Self-heals or escalates — human-approved.

APPLIES TO ANY DEVICE THAT EXPOSES TELEMETRY OR LOGS

Panelboards · Meters · Switchgear · UPS & batteries · PDUs · Transformers · Motors · Breakers

Protecting: hospitals · water & utilities · telecom · manufacturing · data centers · substations

HOW IT WORKS

A closed loop of five agents, guided by Gemini

Gemini Knowledge & Agentic Plane · Gemini 2.5 Flash on Vertex AI

RAG retrieval over device docs · Ask / Plan / Agent assistant · dataset & live-monitor-plan generators

1 · Monitor

Detects anomalies in live & historic telemetry (robust-Z + Isolation Forest).

2 · Diagnose

Root cause via topology correlation + Gemini hypotheses with cited evidence.

3 · Decide

Confidence-gated autonomy — weighs reversibility & asset criticality.

4 · Act

Self-heals reversible faults or hands off to a human.

5 · Verify

Confirms the fix; scores downtime & cost avoided.

Human-in-the-loop approval at Decide → Act

↺ Continuous closed loop — Verify feeds Monitor, so the engine keeps learning · one engine for live + historic

GOOGLE CLOUD & GEN AI STACK

Built end-to-end on Google Cloud



Google Cloud service	Where in Kairos AI	How it is used
Gemini 2.5 Flash · Vertex AI	Ask-the-device, Diagnose agent, generators	Grounded RCA, plain-English answers, dataset & monitor-plan generation
Vertex AI Embeddings (text-embedding-004)	Data Sources · RAG retrieval	Embeds each device's docs into a per-device knowledge base
Firestore	Platform state	Users, devices, sources, chunks & knowledge links — durable across instances
Cloud Storage	Telemetry & datasets	Stores generated/uploaded datasets, SOPs and archives
BigQuery	Audit trail	Partitioned audit table with 30-day retention
Secret Manager	Integrations	GitHub / Confluence keys — never stored in device records
Cloud Run	Deployment	Serverless host; dedicated service account via ADC; scales to zero

RULES CHECK ✅ **Google AI model: Gemini 2.5 Flash** ✅ **Deployed on Cloud Run** ✅ **Agentic 5-agent engine** ✅ **English · functional · scalable**

IMPACT

Hours of downtime → minutes, and lives kept on

Fault (asset)	Path	Down	$ avoided
Voltage sag (Main Incomer)	Auto-resolved	40m	$34,000
Overload (Feeder PDU-2)	Auto-resolved	25m	$11,250
Battery degradation (UPS-3)	Escalated (early)	48m	$40,800
Thermal runaway (UPS-1)	Auto-resolved	55m	$46,750
Harmonic distortion (PDU-1)	Auto-resolved	35m	$15,750
Breaker trip (BRK-4)	Auto-resolved	20m	$5,000
Total · 6 incidents	**5 auto / 1 esc.**	**223m**	**$153,550**

Illustrative synthetic scenarios — not realized savings. Rates: critical $850/min, high $450/min, medium $250/min. MTTR drops from tens of hours to 1–2 cycles.

WHY IT MATTERS

Essential services uptime

Hospitals, water & telecom stay running.

Expertise for everyone

Senior-grade diagnosis without a specialist on site.

Safety

Early detection of fire/overload hazards prevents accidents.

Sustainability

Cuts wasted energy, extends equipment life, fewer truck-rolls.

WHY KAIROS AI WINS

✓ **Deep, real Google Cloud AI**
— Gemini on Vertex AI, embeddings, Firestore, Cloud Storage, BigQuery, Secret Manager, Cloud Run.

✓ **One engine, live + historic**
— The same five-agent brain diagnoses a dataset and a live stream.

✓ **The agent decides what to watch**
— It reads the datasheet to pick parameters and decode event logs.

✓ **Honest by design**
— Offline fallbacks labeled; demo data synthetic; humans approve actions.

Start with one device.

Scale to the world.

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121 automated tests · deployed & verified on Cloud Run · Firestore-backed · Gemini on Vertex AI